Servo General Series

CANOPEN Communication

User Manual V1.0

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CONTENT

Introduction

A series of servo drives that support the standard CANopen communication protocol, allowing Canopen communication parameters to be set via CANopen bus. All parameters of the drive can be controlled via CANopen. In CANopen control mode, the motor's speed, position, torque, start and stop settings can be driven. The servo drive can only be used as a slave in the CANopen bus network. All parameters, parameter values and functions are accessed and accessed through the address composed of index and sub-index.

LD_CANOPEN The internal parameters of the servo drive have realized all parameter mapping, and fully comply with the DS301 standard and part of the DS402 protocol. And it has been widely used in Beckhoff embedded master station EL6751, Siemens PLC, Trio PLC and many other PLC equipment.

1. Introduction to LD_CANOPEN protocol

1.1 Baud rate

The CANOPEN communication baud rate of LD servo drive is preset to 500k bit/s at the factory. Of which baud rate The height below the communication length is related to the communication medium. See the table below for details.

Baud rate	Maximum transmission distance	10BE parameter value
1M	40m	0
500k	130m	1
250k	270m	2
125k	530m	3
50k	1300m	4
20k	3300m	5

1.2 Identifier allocation equipment

The ID range is 1-127, and the default node ID of the drive is 0x08;

Object	COD-Ids	Index of communication parameters in OD
SYNC	80h	1005h
Emergency	80h+node address	1014h
Tx-PDO1	180h+node address	1800h
Rx-PDO1	200h+node address	1400h
Tx-PDO2	280h+node address	1400h
Rx-PDO2	300h+node address	1401h
Tx-PDO3	380h+node address	1802h
Rx-PDO3	400h+node address	1402h
Tx-PDO4	480h+node address	1803h
Rx-PDO4	500h+node address	1403h
Tx-SDO	580h+node address	1200h
Rx-SDO	600h+node address	1200h
NMT Node Guarding	0x700 + Node_ID	1200h

1.3 Introduction to process data object (PDO) communication

Currently, it supports a maximum of 4 RPDO and TPDO, and the relevant parameters comply with the DS301 standard.

PDO communication configuration, each PDO contains three communication parameters: ID number, transmission type and prohibit time. Transmission type press

Currently, it supports a maximum of 4 RPDO and TPDO, and the relevant parameters comply with the DS301 standard.

PDO communication configuration, each PDO contains three communication parameters: ID number, transmission type and prohibition time. According to the CANOPEN protocol, the transmission types are as follows:

--Synchronize:

0: Synchronous message. For RPDO, when PDO data is received, the data will not be updated immediately, but will be updated only when the synchronized message data is received. For TPDO, data will only be sent when a synchronization message is received.

1-240: Synchronization message, (periodically) For RPDO, the data will only be updated after receiving (1 ~ 240) synchronization message (SYNC). For TPDO, the TPDO data will be sent after receiving (1 ~ 240) synchronization message (SYNC). 252: The request command from the remote frame updates the PDO synchronously and responds to the request (not supported temporarily).

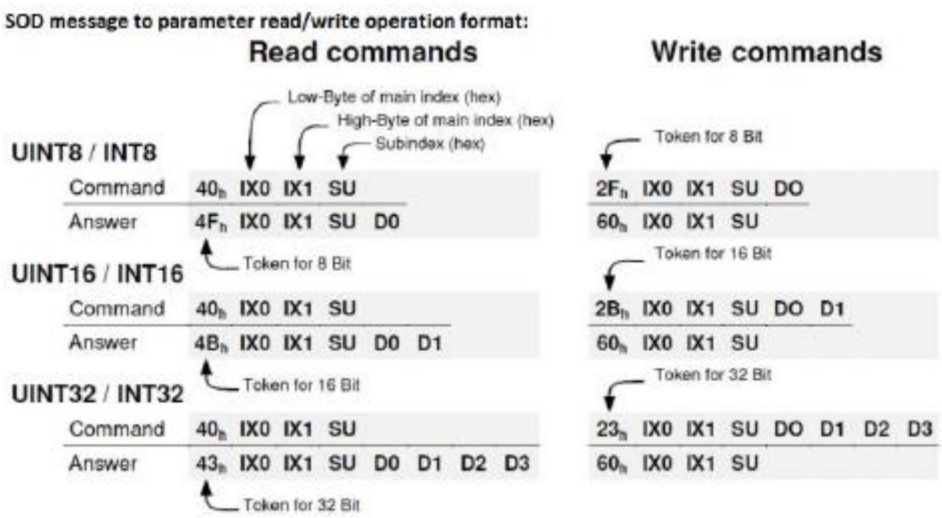
--asynchronous:

253: PDO is continuously updated, but only triggered after remote frame request (not supported temporarily).

255: Asynchronous (for RxPDO, it will be updated every time data is received. For TxPDO, it will be sent when the content changes and exceeds the prohibition time). The prohibition time is only applicable to TxPDO, and the default is 100 microseconds.

LD_CANOPEN servo drive adopts synchronous message mode, and the default transmission type of RxPDO and TxPDO are both 255.

1.4 Service data object (SDO) communication protocol



For example:

UINT8 / INT8	Reading of Obj. 6061_00_h Returning data: 01_h	Writing of Obj. 1401_02_h Data: EF_h
Command:	40_h 61_h 60_h 00_h	$2F_h$ 01_h 14_h 02_h EF_h
Answer:	$4F_h$ 61_h 60_h 00_h 01_h	60_h 01_h 14_h 02_h
UINT16 / INT16	Reading of Obj. 6041_00_h Returning data: 1234_h	Writing of Obj. 6040_00_h Data: $03E8_h$
Command:	40_h 41_h 60_h 00_h	$2B_h$ 40_h 60_h 00_h $E8_h$ 03_h
Answer:	$4B_h$ 41_h 60_h 00_h 34_h 12_h	60_h 40_h 60_h 00_h
UINT32 / INT32	Reading of Obj. 6093_01_h Returning data: 12345678_h	Writing of Obj. 6093_01_h Data: 12345678_h
Command:	40_h 93_h 60_h 01_h	23_h 93_h 60_h 01_h 78_h 56_h 34_h 12_h
Answer:	43_h 93_h 60_h 01_h 78_h 56_h 34_h 12_h	60_h 93_h 60_h 01_h

SOD error message format

Command:	...	IX0	IX1	SU	...	...	...	...
Answer:	80_h	IX0	IX1	SU	F0	F1	F2	F3
	Error token				Error code (4 Byte)			

Error code F3 F2 F1 F0	Description
05 03 00 00_h	Toggle bit not alternated
05 04 00 01_h	Client / server command specifier not valid or unknown
06 01 00 00_h	Unsupported access to an object
06 01 00 01_h	Attempt to read a write only object
06 01 00 02_h	Attempt to write a read only object
06 02 00 00_h	Object does not exist in the object dictionary
06 04 00 41_h	Object cannot be mapped to the PDO
06 04 00 42_h	The number and length of the objects to be mapped would exceed PDO length
06 04 00 47_h	General internal incompatibility in the device
06 07 00 10_h	Data type does not match, length of service parameter does not match
06 07 00 12_h	Data type does not match, length of service parameter too high
06 07 00 13_h	Data type does not match, length of service parameter too low
06 09 00 11_h	Sub-index does not exist
06 04 00 43_h	General parameter incompatibility
06 06 00 00_h	Access failed due to an hardware error *1)
06 09 00 30_h	Value range of parameter exceeded
06 09 00 31_h	Value of parameter written too high
06 09 00 32_h	Value of parameter written too low
06 09 00 36_h	Maximum value is less than minimum value
08 00 00 20_h	Data cannot be transferred or stored to the application *1)
08 00 00 21_h	Data cannot be transferred or stored to the application because of local control
08 00 00 22_h	Data cannot be transferred or stored to the application because of the present device state *3)
08 00 00 23_h	No Object Dictionary is present *2)

1.5 MNT node protection

LD_CANOPEN series servo drives are configured to generate periodic heartbeat messages (Heartbeat) Message.

Heartbeat Producer → Consumer(s)

COB-ID	Byte 0
0x700 + Node_ID	State

The status can be the following values:

State	Meaning
0	Boot-up
4	Stopped
5	Operational
127(0x7F)	Pre-operational

When a Heartbeat node starts, its Boot-up message is its first Heartbeat message. Heartbeat consumers are usually NMT-Master nodes, which set a timeout value for each Heartbeat node. Take corresponding actions when the timeout occurs.

1.6 Emergency

Every time a fault is detected, an emergency signal will be thrown out and stored in the 1003 field

The COB-ID of the emergency object is 0x81 to 0xFF. EMCY objects are fully defined in CiA DS 301. The structure of manufacturer-specific emergency messages is as follows:

0	Error code
1	
2	Error register
3	Internal error code
4	Error code data field 1
5	
6	Error code data field 2
7	

Fault code table:

630F-01-00-0000-0000-Missing parameter failure (memory loading)

7110-01-00-0000-0000-power level fault

1000-01-00-0000-0000-General fault 1 (fault lock indication)

2211-03-00-0000-0000-overcurrent

2212-03-00-0000-0000-Current is out of tolerance

2221-03-00-0000-0000-Continuous overcurrent (overload)

2310-03-00-0000-0000-current fault 1 (OCP)

3130-01-00-0000-0000-Motor phase loss

3220-05-00-0000-0000-Voltage fault 1 (under voltage)

3210-05-00-0000-0000-Voltage fault 2 (overvoltage)
4210-08-00-0000-0000-Drive is overheated
7305-01-00-0000-0000-Encoder abnormal
7308-01-00-0000-0000- Hall feedback signal is abnormal
7311-01-00-0000-0000-speed out of tolerance
7320-01-00-0000-0000-Position abnormal
8131-01-00-0000-0000-Communication timeout (off enable)
8611-01-00-0000-0000-Position following abnormal
FF00-80-00-0000-0000-Excessive acceleration warning
FF01-80-00-0000-0000-Acceleration too small warning
FF02-80-00-0000-0000-IGBT failure
FF03-80-00-0000-0000-Motor overheated
FF04-80-00-0000-0000-speed failure
FF08-80-00-0000-0000-Internal zero offset fault

Note: unused bytes must be set to zero

2. Process data object (PDO) mapping

2.1 The default parameters of the driver

The default parameter TXPDO of the driver is a 100ms transmission cycle, and RXPDO is an asynchronous reception. The mapping is as follows:

TXPDO 0

Subindex	Value (H)	Drive internal parameter name
0	4	_
1	60410010	Status word
2	200B0003	Real-time failure 1
3	200B0004	Real-time status 1
4	60610008	Display Mode

TXPDO 1

Subindex	Value (H)	Drive internal parameter name
0	2	_
1	60640020	Current position feedback value
2	60690020	Current speed feedback value

TXPDO 2

Subindex	Value (H)	Drive internal parameter name
0	2	_
1	60640020	Expected target location
2	60690020	Expected target speed

TXPDO 0

Subindex	Value (H)	Drive internal parameter name
0	2	_
1	60400010	Control Word
2	60600008	Operation Mode

TXPDO 1

Subindex	Value (H)	Drive internal parameter name
0	2	_
1	60640020	Target Position
2	60690020	Profile Velocity

TXPDO 2

Subindex	Value (H)	Drive internal parameter name
0	1	_

1	60FF0020	Speed setting of profile speed loop

2.2. Illustrate RXPDO mapping

If the RXPDO mapping of the default parameters of the drive is configured, assume that the node id is 1

2.2.1 Configure RXPDO 0

(1) Do not enable RXPDO 0 to receive first

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	22	00	14	01	01	02	00	80

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	00	14	01	00	00	00	00

(2), clear RXPDO 0 mapping

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	2F	00	16	00	00	00	00	00

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	00	16	00	00	00	00	00

(3) Configure the mapping of sub-index 01 of RXPDO 0 to 60400010, which is the control word.

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	22	00	16	01	10	00	40	60

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	00	16	01	00	00	00	00

(4) Configure the mapping of RXPDO 0 with sub-index 02 to 60600008, which is the working mode.

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	22	00	16	02	08	00	60	60

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	00	16	02	00	00	00	00

(5) The configuration of RXPDO 0 is asynchronous trigger mode.

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	2F	00	14	02	FF	00	00	00

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	00	14	02	00	00	00	00

(6) Set the number of objects mapped by RXPDO 0 to 2

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	2F	00	16	00	02	00	00	00

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	00	16	00	00	00	00	00

(7) Enable RXPDO 0 to receive

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	22	00	14	01	01	02	00	00

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	00	14	01	00	00	00	00

2.2.2 Configure RXPDO 1

(1) Do not enable RXPDO 1 reception first

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	22	01	14	01	01	03	00	80

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	01	14	01	00	00	00	00

(2) Clear RXPDO 1 mapping

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	2F	01	16	00	00	00	00	00

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	01	16	00	00	00	00	00

(3) Configure the mapping of the sub-index of RXPDO 1 as 01 to 607A0020, which is the given position.

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	22	01	16	01	20	00	7A	60

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	01	16	01	00	00	00	00

(4) Configure the mapping of RXPDO 1 sub-index 02 to 60810020, which is the working mode.

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	22	01	16	02	20	00	81	60

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	01	16	02	00	00	00	00

(5) The configuration of RXPDO 1 is asynchronous trigger mode.

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	2F	01	14	02	FF	00	00	00

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	01	14	02	00	00	00	00

(6) Set the number of objects mapped by RXPDO 1 to 2

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	2F	01	16	00	02	00	00	00

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	01	16	00	00	00	00	00

(7) Enable RXPDO 1 to receive

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	22	01	14	01	01	03	00	00

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	01	14	01	00	00	00	00

2.3 Illustrate TXPDO mapping

If you configure the TXPDO mapping of the drive's default parameters, assume that the node id is 1.

2.3.1 Configure TXPDO 0

(1) Disable TXPDO 0 to receive first

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	22	00	18	01	81	01	00	80

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	00	18	01	00	00	00	00

(2) Clear TXPDO 0 mapping

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	2F	00	1A	00	00	00	00	00

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	00	1A	00	00	00	00	00

(3) Configure the mapping of the sub-index of TXPDO 0 as 01 to 60410010, which is the status word.

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	22	00	1A	01	10	00	41	60

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	00	1A	01	00	00	00	00

(4) Configure the mapping with the sub-index of TXPDO 0 as 02 to 200B0310, that is, real-time fault 1.

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	22	00	1A	02	10	03	0B	20

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	00	1A	02	00	00	00	00

(5) Configure the mapping with the sub-index of TXPDO 0 as 03 as 200B0410, that is, real-time status 1.

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	22	00	1A	03	10	04	0B	20

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	00	1A	03	00	00	00	00

(6) Configure the mapping of TXPDO 0's sub-index 04 to 60610008, that is, display the working mode.

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	22	00	1A	04	08	00	61	60

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	00	1A	04	00	00	00	00

(7) Set the number of objects mapped by TXPDO 0 to 4

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	2F	00	1A	00	03	00	00	00

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	00	1A	00	00	00	00	00

(8) The configuration of TXPDO 0 is asynchronous trigger mode.

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	2F	00	18	02	FF	00	00	00

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	00	18	02	00	00	00	00

(9) Configure the suppression time bit of TXPDO 0 to 10ms.

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	2B	00	18	03	64	00	00	00

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	00	18	03	00	00	00	00

(10) Configure the trigger time bit of TXPDO 0 to 10ms.

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	2B	00	18	05	0A	00	00	00

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	00	18	05	00	00	00	00

(11), enable TXPDO 0 to receive

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	22	00	18	01	81	01	00	00

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	00	18	01	00	00	00	00

2.3.2 Configure TXPDO 1

(1) Disable TXPDO 1 reception first

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	22	01	18	01	81	02	00	80

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	00	18	01	00	00	00	00

(2) Clear TXPDO 1 mapping

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	2F	01	1A	00	00	00	00	00

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	01	1A	00	00	00	00	00

(3) Configure the mapping with the sub-index 01 of TXPDO 1 to be 60640020, which is the current position feedback value.

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	22	01	1A	00	00	00	00	00

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	01	1A	01	00	00	00	00

(4) Configure the mapping with the sub-index of TXPDO 1 as 02 to 6069020, which is the current speed feedback value.

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	22	01	1A	02	20	00	69	60

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	01	1A	02	00	00	00	00

(5) Set the number of objects mapped by TXPDO 1 to 2

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7

601	2F	01	1A	00	02	00	00	00

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	01	1A	00	00	00	00	00

(6) The configuration of TXPDO 1 is asynchronous trigger mode.

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	2F	01	18	02	FF	00	00	00

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	01	18	02	00	00	00	00

(7) Configure the suppression time bit of TXPDO 1 to 10ms.

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	2B	00	18	03	64	00	00	00

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	01	18	03	00	00	00	00

(8) Configure the trigger time bit of TXPDO 1 to 10ms.

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	2B	01	18	05	0A	00	00	00

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	01	18	05	00	00	00	00

(9) Enable TXPDO 1 reception

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	22	01	18	01	81	02	00	00

Driver response

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
581	60	01	18	01	00	00	00	00

3. Introduction of CANOPEN servo drive startup process

If the user controls the company's servo drive through the CANOPEN interface, the control method follows the DS402 protocol.

The proposed control method.

First, after the LD_CANOPEN servo drive is powered on, it will automatically enter the initialization state (0x00), and will report to the master station

Send a heartbeat message, and then enter the pre-operation state (0x7F), and wait for the response of the master to control the node state of the drive.

Only when the slave station enters the operating state (0x05), it enters the state of process data operation, and the drive will report the process regularly

Data, and the master station can also control the drive through process data.

Since the control methods are all standard control methods and have been successfully controlled in the Beckhoff main station CX9010-0001, it can be

The user provides an eds file description, it is not necessary to introduce it in detail here.

That is, the master controller only needs to send 2 bytes of data to id: 0x00: 01 02 (the first byte represents the command, the second

Each byte represents the node address), then the normal operation mode can be entered, that is, PDO communication can be carried out.

(The master controller only needs to send 2 bytes of data to id: 0x00: 80 02 (the first byte represents the command, the second word

Section represents the node address), you can enter the pre-operation operation mode, and PDO communication is not allowed)

(The master controller only needs to send 2 bytes of data to id: 0x00: 02 02 (the first byte represents the command, the second word

Section represents the node address), then it can enter the stop mode, and PDO and SDO communication is not allowed)

4. Introduction of CANOPEN communication control servo driver

4.1 Introduction to the control state machine

The LD servo drive supports a state machine. The master station controls the drive through the control word (control word).

Read the status word of the drive to know the current status of the drive. This chapter will use the following Narrative:

State: If the main power is activated or an alarm occurs, the servo drives are in different states. CANOPEN total The state machine under line control will be explained in this section.

For example: SWITCH_ON_DISABLED

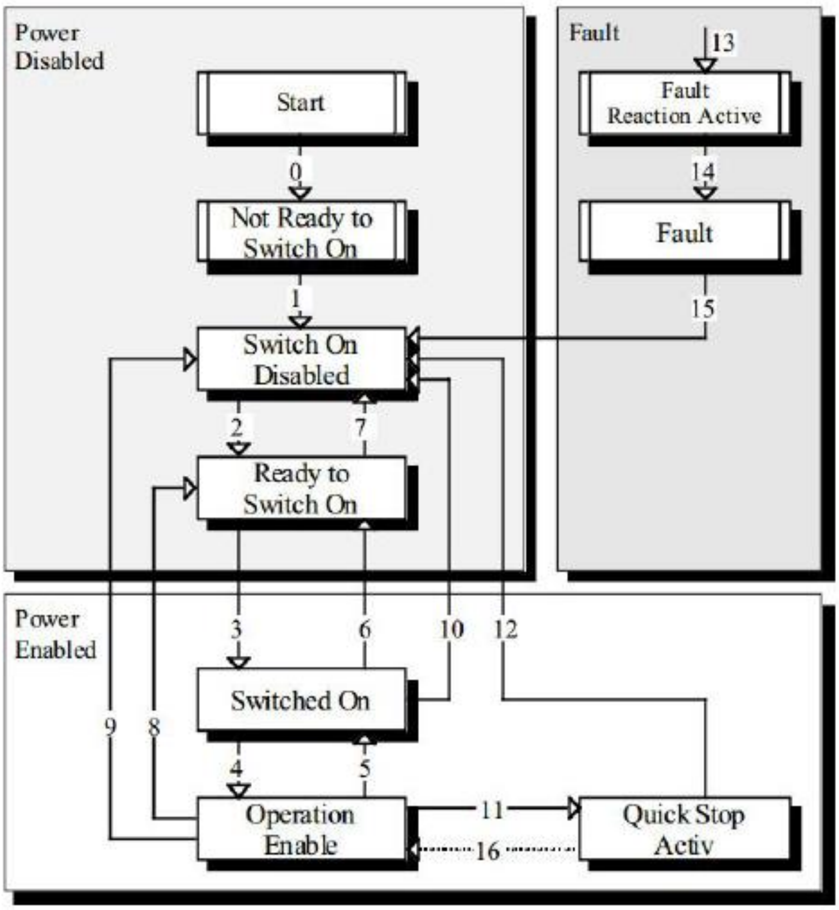
State Transition: The state machine also defines how to transition from one state to another. State

The transfer depends on the console controlled by the master station or the drive itself, for example, the drive generates an alarm.

Command: In order to start State Transition, the bit combination of control word is defined,

These bit combinations are called Commands.

State diagram (state diagram): All State (state) and State Transition (state transmission) constitute the State diagram (state diagram).



As shown in the figure above, the state machine can be divided into three parts: "Power Disabled" (main power off), "Power
Enabled" (main power on) and "Fault". All states enter "Fault" after an alarm occurs.
After power-on, the drive completes initialization and then enters the SWITCH_ON_DISABLED state. In this state, you can
In order to carry out CAN communication, the drive can be configured (for example, the working mode of the drive can be set to
"PP" mode). At this time, the main power is still off and the motor is not excited. After State Transition (State Transition
After entering) 2, 3, 4, enter OPERATION ENABLE. At this time, the main power is turned on, and the drive is
Work mode to control the motor. Therefore, before this state, you must first confirm that the drive parameters and corresponding parameters have been correctly configured.
The input value for is zero. State Transition 9 completes closing the main circuit of the circuit. Once the drive alarms, The status of the drives all enter FAULT.

State name	Description
Not Ready to Switch On	The servo drive is in the process of initializing, and CAN communication is not possible.
Switch On Disabled	The initialization of the servo drive is completed and CAN communication can be carried out.
Ready to Switch On	The servo drive is waiting to enter the Switch On state, and the motor is not excited.
Switched On	The servo drive is ready for servo, and the main power is on.
Operation Enable	The servo drive servo inputs the excitation signal to the motor, and controls the motor according to the control mode.
Quick Stop Active	The servo drive will stop according to the set mode.
Fault Reaction Active	The servo drive detects the occurrence of an alarm and stops according to the set method, the motor still There is an excitation signal.
Fault	The motor has no excitation signal.

4.2, Control word description

Index	6040h
Name	Status Word
Object Code	VAR
Data Type	UINT16
Access	RO
PDO Mapping	YES
Units	_
Value Range	_
Default Value	0

Control Word bit description is shown in the figure below:

15	11	10	9	8	7	6	4	3	2	1	0
Manufacturer specific		Reserved		Halt	Fault reset	Operation mode specific		Enable operation	Quick stop	Enable voltage	Switch on

4.3 Status Word description

Index	6041h
Name	Status Word
Object Code	VAR
Data Type	UINT16
Access	RO
PDO Mapping	YES
Units	_
Value Range	_
Default Value	_

Status Word bit description is shown in the figure below

Bit	Description
0	Ready to switch on
1	Switched on
2	Operation enabled
3	Fault
4	Voltage enable
5	Quick stop
6	Switch on disabled
7	Warning
9~8	Reserved
10	Target reached
11	Internal limit active
13~12	Operation mode specific
15~14	Reserved

4.4. Introduction to control methods

LD servo drive currently supports 4 control modes in CANOPEN DSP402:

Homing mode

Speed control mode

Current control mode

Position control mode

Periodic synchronous position mode

4.4.1, control mode related parameters

Index	Object	Name	Type	Attr.
6060 h	VAR	Operation Mode	INT8	RW
6061 h	VAR	Display Mode	INT8	RO

4.4.2、Operation Mode

The control mode of the servo drive is determined by the Operation Mode parameter.

Index	6060h
Name	Operation Mode
Object Code	VAR
Data Type	INT8
Access	RW
PDO Mapping	YES
Units	--
Value Range	1、3、4、6、8
Default Value	--

Value description of Operation Mode:

Value	**Description**
0	Unset control mode
1	Position control mode
3	Speed control mode
4	Current control mode
8	Periodic synchronous position mode

4.4.3、Display Mode

The current control mode of the servo drive can be known by reading the Display Mode parameter.

Index	6061h
Name	Display Mode
Object Code	VAR
Data Type	INT8
Access	RO
PDO Mapping	YES
Units	--
Value Range	1、3、4、6、8
Default Value	--

Note:

1. The current control mode of the drive can only be known through the Display Mode parameter;
2. Only when Target Reached state, the drive control mode will switch from the current mode to the set control mode. Control mode, Display Mode can be consistent with Operation Mode.

5. Return to zero mode

The LD servo drive currently supports a variety of zero return methods, and the user should choose the corresponding zero return method reasonably. For example: If the motor uses an incremental encoder, you can choose the zero return method by C pulse; if the motor uses a serial encoder or resolver, you cannot choose the zero return method by C pulse.

The user can set the zero return method and the zero return speed. After the driver finds the reference point position, it can set the zero position to offset the reference point Home Position (607C h) forward.

5.1 Control word of return to zero mode

15~9	8	7~5	4	3~0
*	Halt	*	home_start_operation	*

Name	Value	Description
Homing operation start	0	Homing mode inactive
	0 → 1	Start homing mode
	1	Homing mode active
	1 → 0	Interrupt homing mode
Halt	0	Execute the instruction of bit 4
	1	Stop axle with homing acceleration

5.2 The status word of the zero return mode

15 ~ 14	13	12	11	10	9~0
*	homing error	homing attained	*	target reached	*

Name	Value	Description
Target reached	0	Halt = 0: Home position not reached Halt = 1: Axle decelerates
	1	Halt = 0: Home position reached Halt = 1: Axle has velocity 0
Homing attained	0	Homing mode not yet completed
	1	Homing mode carried out successfully
Homing error	0	No homing error
	1	Homing error occurred; Homing mode carried out not successfully; The error cause is found by reading the error code

5.3 Relevant parameters of zero return mode

Index	Object	Name	Type	Attr..
607C h	VAR	Home Offset	INT32	RW
6098 h	VAR	Homing Method	INT8	RW
6099 h	ARRAY	SHOM_Speed	UINT32	RW

5.3.1 Object 607Ch: Home offset

The zero return offset is the deviation between the application program and the mechanical zero point.

Index	**607Ch**
Name	Home Offset
Object Code	VAR
Data Type	INT32
Access	RW
PDO Mapping	Possible
Units	Number of pulses
Value Range	INT32
Default Value	0

5.3.2 Object 6098h: Homing method

Index	6098h
Name	Homing method
Object Code	VAR
Data Type	INT8
Access	RW
PDO Mapping	Possible
Units	_
Value Range	INTEGER8
Default Value	34

5.3.3 Object 6099h: Homing speeds

Sub-Index	**0**
Description	Number of entries
Entry Category	Mandatory
Access	ro
PDO Mapping	No
Value Range	2

Default Value	2

Sub-Index	**1**
Description	Speed during search for switch
Entry Category	Mandatory
Access	rw
PDO Mapping	Possible
Value Range	UNSIGNED32
Default Value	100

Sub-Index	**2**
Description	Speed during search for zero
Entry Category	Mandatory
Access	rw
PDO Mapping	Possible
Value Range	UNSIGNED32
Default Value	100

6. Speed control mode

6.1 Control word of speed mode

15~9	8	7~4	3~0
*	Halt	*	*

Name	Value	Description
Halt	0	Execute the motion
	1	Stop axle

6.2 Status word of speed mode

15 ~ 14	13	12	11	10	9~0
*	Max Slippage Error	Speed	*	target reached	*

Name	Value	Description
Target reached	0	Halt = 0: *Target velocity* not (yet) reached Halt = 1: Axle decelerates
	1	Halt = 0: *Target velocity* reached Halt = 1: Axle has velocity 0
Speed	0	Speed is not equal 0
	1	Speed is equal 0
Max slippage error	0	Maximum slippage not reached
	1	Maximum slippage reached

6.3 Related parameters of speed control mode

Index	Name
6069h	Velocity Sensor Actual Value
606Ah	Sensor Select
606Bh	Velocity Demand Value
606Ch	Velocity Actual Value
606D h	Velocity window
606E h	Velocity window time

606F h	Velocity threshold
6070 h	Velocity threshold time
60FF h	Target Velocity

6.3.1 Object 6069h: Velocity sensor actual value

Index	6069h
Name	Velocity sensor actual value
Object Code	VAR
Data Type	INT32
Access	RO
PDO Mapping	Possible
Units	RPM
Value Range	INT32
Default Value	0

6.3.2 Object 606Ah: Sensor selection code

Index	606Ah
Name	Sensor selection code
Object Code	VAR
Data Type	INT16
Access	RO
PDO Mapping	Possible
Units	---
Value Range	INT16
Default Value	0

6.3.3 Object 606Bh: Velocity demand value

Index	606Bh
Name	Velocity demand value
Object Code	VAR
Data Type	INT32
Access	RO
PDO Mapping	Possible
Units	RPM
Value Range	INT32
Default Value	0

6.3.4 Object 606Ch: Velocity actual value

Index	606Ch
Name	Velocity actual value
Object Code	VAR
Data Type	INT32
Access	RO
PDO Mapping	Possible
Units	RPM
Value Range	INT32
Default Value	0

6.3.5 Object 606Dh: Velocity window

Index	606Dh
Name	Velocity window
Object Code	VAR
Data Type	UINT16
Access	RW
PDO Mapping	Possible
Units	RPM
Value Range	UINT16
Default Value	0

6.3.6 Object 606Eh: Velocity window time

Index	606Eh
Name	Velocity window time
Object Code	VAR
Data Type	UINT16
Access	RW
PDO Mapping	Possible
Units	RPM
Value Range	UINT16
Default Value	0

6.3.7 Object 606Fh: Velocity threshold

Index	606Fh
Name	Velocity threshold

Object Code	VAR
Data Type	UINT16
Access	RW
PDO Mapping	Possible
Units	RPM
Value Range	UINT16
Default Value	0

6.3.8 Object 6070h: Velocity threshold time

Index	6070h
Name	Velocity threshold time
Object Code	VAR
Data Type	UINT16
Access	RW
PDO Mapping	Possible
Units	RPM
Value Range	UINT16
Default Value	0

6.3.9 Object 60FFh: Target velocity

Index	60FFh
Name	Target velocity
Object Code	VAR
Data Type	INT32
Access	RW
PDO Mapping	Possible
Units	RPM
Value Range	INT32
Default Value	0

7.Current control mode

7.1 Status word of current mode

15 ~ 14	13	12	11	10	9~0
*	*	*	*	target reached	*

Name	Value	Description
Target reached	0	Halt = 0: *Target velocity* not (yet) reached Halt = 1: Axle decelerates
	1	Halt = 0: *Target velocity* reached Halt = 1: Axle has velocity 0

7.2. Related parameters of current control mode

Index	Name
6071h	Target Torque
6072h	Max torque
6073h	Max current
6074h	Torque demand value
6075h	Motor rated current
6076h	Motor rated torque
6077h	Torque actual value
6078h	Current actual value
6087h	Torque slope

7.2.1 Object 6071h: Target torque

Index	6071h
Name	Target torque
Object Code	VAR
Data Type	INT16
Access	RW
PDO Mapping	Possible
Units	0.1 amp
Value Range	INT16
Default Value	0

7.2.2 Object 6072h: Max torque

Index	6072h
Name	Max torque
Object Code	VAR
Data Type	UINT16
Access	RO
PDO Mapping	Possible
Units	---
Value Range	UINT16
Default Value	0

7.2.3 Object 6073h: Max current

Index	6073h
Name	Max current
Object Code	VAR
Data Type	UINT16
Access	RW
PDO Mapping	Possible
Units	0.1 amp
Value Range	UINT16
Default Value	0

7.2.4 Object 6074h: Torque demand value

Index	6074h
Name	Torque demand value
Object Code	VAR
Data Type	INT16
Access	RO
PDO Mapping	Possible
Units	0.1 amp
Value Range	INT16
Default Value	0

7.2.5 Object 6075h: Motor rated current

Index	6075h
Name	Motor rated current
Object Code	VAR

Data Type	UINT32
Access	RO
PDO Mapping	Possible
Units	---
Value Range	UINT32
Default Value	0

7.2.6 Object 6076h: Motor rated torque

Index	6076h
Name	Motor rated torque
Object Code	VAR
Data Type	UINT32
Access	RO
PDO Mapping	Possible
Units	---
Value Range	UINT32
Default Value	0

7.2.7 Object 6077h: Torque actual value

Index	6077h
Name	Torque actual value
Object Code	VAR
Data Type	INT16
Access	RO
PDO Mapping	Possible
Units	0.1 amp
Value Range	INT16
Default Value	0

7.2.8 Object 6078h: Current actual value

Index	6078h
Name	Current actual value
Object Code	VAR
Data Type	INT16
Access	RO
PDO Mapping	Possible
Units	0.1 amp
Value Range	INT16
Default Value	0

7.2.9 Object 6087_h: Torque slope

Index	6087h
Name	Torque slope
Object Code	VAR
Data Type	UINT32
Access	RO
PDO Mapping	Possible
Units	---
Value Range	UINT32
Default Value	0

8. Position control mode

8.1. Control word of position mode

15~9	8	7	6	5	4	3~0
*	Halt	*	asb/rel	Change set immediately	New set point	*

Name	Value	Description
New set-point	0	Does not assume target position
	1	Assume target position
Change set immediately	0	Finish the actual position and then start the next positioning
	1	Interrupt the actual positioning and start the next positioning
asb/rel	0	Target the position is an absolute value
	1	Target the position is an relative value
Halt	0	Execute positioning
	1	Stop axle with profile deceleration(if not supported with profile acceleration)

8.2. Status word of position mode

15~9	13	12	11	10	9~0
*	Following error	Set-point acknowledge	*	Target reached	*

Name	Value	Description
Target reached	0	Halt=0: Target position not reached Halt=1: Alex decelerates
	1	Halt=0: Target position reached Halt=1: Velocity of axle is 0
Set-point acknowledge	0	Trajectory generator has not assumed the positioning values(yet)
	1	Trajectory generator has assumed the positioning values(yet)
Following error	0	Target the position is an absolute value
	1	Not following error
	1	Following error

8.3. Position control related parameters

Index	Name
6064h	Position actual value
607Ah	Target Position
607Bh	Position range limit
607Dh	Software position limit
607Fh	Maximum profile velocity
6081 h	Profile velocity
6082 h	End velocity
6083 h	Profile acceleration
6084 h	Profile deceleration
6086 h	Motion-profile-type
60C5 h	Maximum acceleration
60C6 h	Maximum deceleration
60F2 h	Positioning option code

8.3.1 Object 6064$_h$: Position actual value

Index	6064h
Name	Position actual value
Object Code	VAR
Data Type	INT32
Access	RO
PDO Mapping	YES
Units	Position units
Value Range	----
Default Value	----

8.3.2 Object 607A$_h$: Target position

Index	607Ah
Name	Position actual value
Object Code	VAR
Data Type	INT32
Access	RW
PDO Mapping	YES
Units	Position units
Value Range	----
Default Value	0

8.3.3 Object 607B$_h$: Position range limit

Sub-Index	0
Description	Number of entries
Access	Mandatory
PDO Mapping	ro
Access	No
Value Range	2
Default Value	2

Sub-Index	1
Description	Min position range limit
Access	Mandatory
PDO Mapping	rw
Access	No
Value Range	INT32
Default Value	-2^{31}

Sub-Index	2
Description	Max position range limit
Access	Mandatory
PDO Mapping	rw
Access	No
Value Range	INT32
Default Value	$2^{31}-1$

8.3.4 Object 607D$_h$: Software position limit

Sub-Index	0
Description	Number of entries
Access	Mandatory
PDO Mapping	ro
Access	No
Value Range	2
Default Value	-2^{31}

Sub-Index	1
Description	Min position limit
Access	Mandatory
PDO Mapping	rw
Access	No

Value Range	INT32
Default Value	2_{31}-1

Sub-Index	2
Description	Max position limit
Access	Mandatory
PDO Mapping	rw
Access	No
Value Range	INT32
Default Value	0

8.3.5 Object 6081h: profile velocity

Profile velocity refers to the speed finally reached after the position is started and the acceleration is completed. (Unit: RPM)

Index	6081h
Name	Profile velocity
Object Code	VAR
Data Type	UINT32
Access	RW
PDO Mapping	YES
Units	Speed units
Value Range	----
Default Value	0

8.3.6 Object 6082h: end velocity

End velocity is the velocity when reaching a given position (target position). Usually in order to stop the drive when reaching a given position, this parameter is set to 0; but in continuous multi-point position, this value can be set to a non-zero value, and

The default is the same size as profile velocity. (Unit: RPM)

Index	6082h
Name	End velocity
Object Code	VAR
Data Type	UINT32
Access	RW
PDO Mapping	YES
Units	Speed units
Value Range	----
Default Value	0

8.3.7 Object 6083_h: profile acceleration

Profile acceleration is the acceleration during reaching a given position.

Index	6083h
Name	Profile acceleration
Object Code	VAR
Data Type	UINT32
Access	RW
PDO Mapping	YES
Units	Acceleration units
Value Range	----
Default Value	2000RPM/s

8.3.8 Object 6084h: profile deceleration

Profile deceleration is the deceleration during reaching a given position. (Not currently supported)

Index	6084h
Name	Profile acceleration
Object Code	VAR
Data Type	UINT32
Access	RW
PDO Mapping	YES
Units	Acceleration units
Value Range	----
Default Value	6000RPM/s

8.3.9 Object 6086_h: Motion profile type

Motion_profile_type is used to select which speed profile. Only trapezoidal speed curve is supported in the current position. Write 0 to select the trapezoidal speed curve. (Currently only supports trapezoidal speed curve)

Index	6086h
Name	Motion_profile_type
Object Code	VAR
Data Type	INT16
Access	RW
PDO Mapping	YES
Units	----
Value Range	0
Default Value	0

8.3.10 Object 60C5h: Max acceleration

Index	60C5h
Name	Max acceleration
Object Code	VAR
Data Type	UINT32
Access	RW
PDO Mapping	Possible
Units	----
Value Range	0
Default Value	0

8.3.11 Object 60C6h: Max deceleration

Index	60C6h
Name	Max deceleration
Object Code	VAR
Data Type	UINT32
Access	RW
PDO Mapping	Possible
Units	----
Value Range	0
Default Value	0

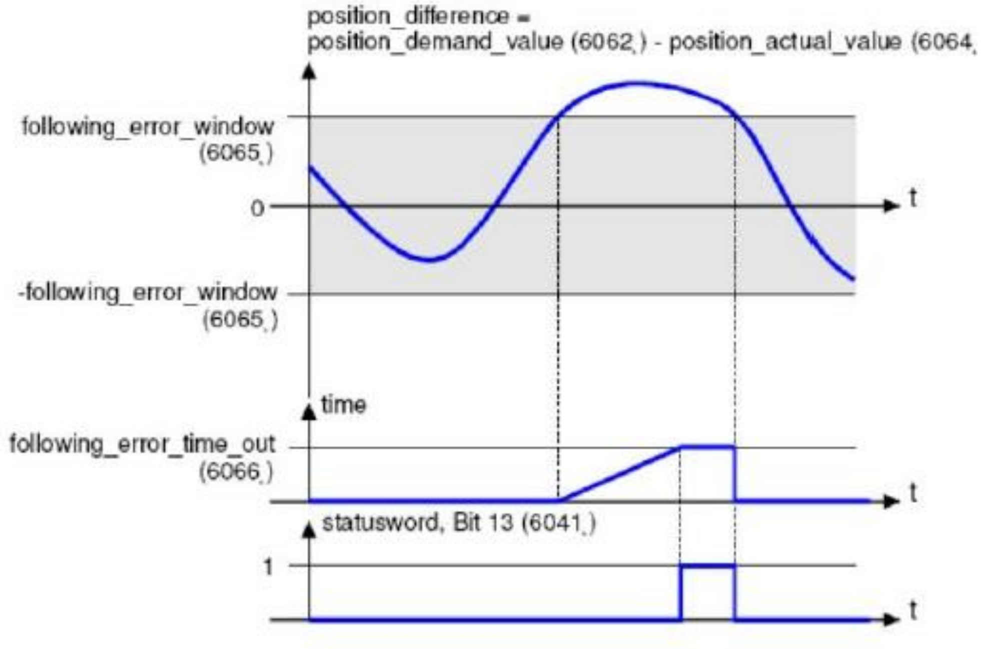
8.4. Position control function

This chapter mainly describes the parameters required in the position control mode. The desired position output by the Trajectory unit is used as the input of the drive position loop. In addition, the actual position is measured by the motor encoder. The action of the position controller is affected by the parameter settings. In order to control the stability of the system.The output value of the position loop must be limited. The output value is used as the given speed of the speed loop. In Factor

Group unit, all input and output values are converted into the corresponding internal unit of the drive.

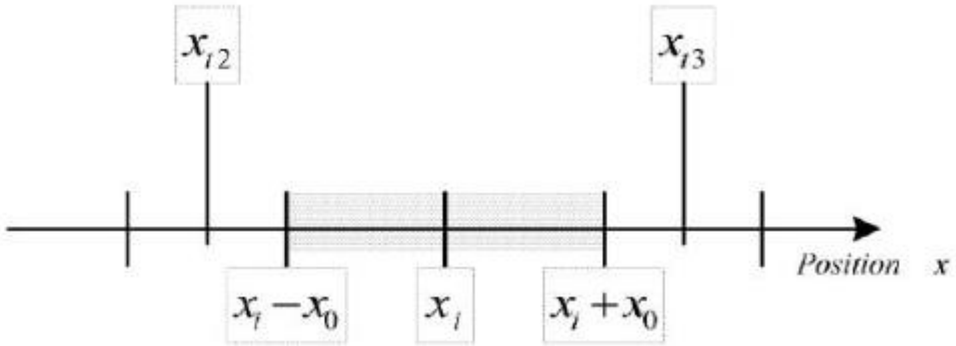
The following describes the sub-functions of position control:

8.4.1 Following error



Following error -function description

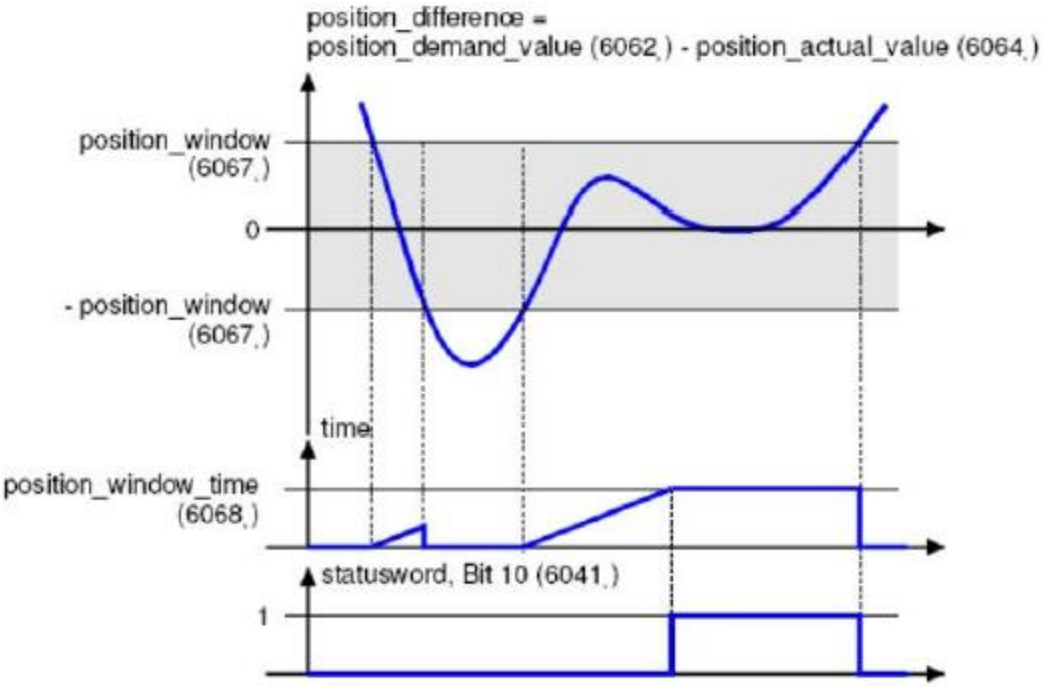
The following error refers to the deviation between the actual position and the desired position. As shown in the figure above, if the following error value is greater than the following error window within the set time, then bit13 (following error) of the status word will be set to 1.



Following error-for example

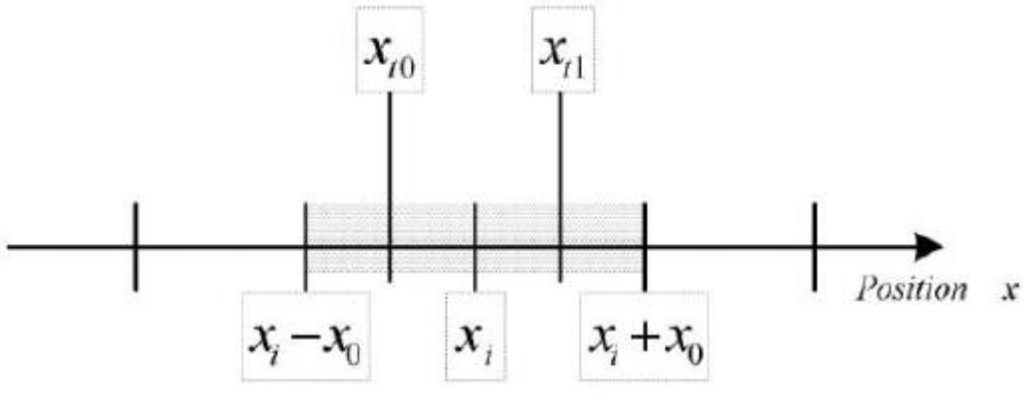
8.4.2 Position reached

This function defines the location window near the target location. If the actual position of the drive is stable within this range (position window) for the set time, then the bit10 (target reached) of the status word will be set to 1. As shown below.



Location arrival-functional description

The following figure shows that the position window is symmetrically distributed near the target position, that is, the range of xi-x0 to xi+x0. For example, the positions xt0 and xt1 are in the position window. If the drive is in the window, a certain period of time starts to count. If the fixed period reaches the set value (position_window_time) and the drive position is always in the window during this period, the bit10 (target reached) of the status word will be set to 1. As soon as the drive position leaves the window, bit10 (target reached) of the status word will be cleared immediately



Location reached-example

8.5 Position control related parameters

Index	Object	Name	Type	Attr.
6062 h	VAR	position_demand_value	INT32	RO
6063h	VAR	position_actual_value*	INT32	RO
6064h	VAR	position_actual_value	INT32	RO
6065h	VAR	following_error_window	UINT32	RW
6066h	VAR	following_error_time_out	UINT16	RW
6067h	VAR	position_window	UINT32	RW
6068h	VAR	position_time	UINT16	RW
60FA	VAR	Control effort	INT32	RO

Index	6062h
Name	Position_demand_value
Object Code	VAR
Data Type	INT32
Access	RO
PDO Mapping	YES
Units	Position units
Value Range	----
Default Value	----

Index	6063h
Name	Position_ actual _value
Object Code	VAR
Data Type	INT32
Access	RO
PDO Mapping	YES
Units	Position units
Value Range	----
Default Value	----

Index	6065h
Name	Following_error_window
Object Code	VAR
Data Type	INT32
Access	RW
PDO Mapping	YES
Units	Position units
Value Range	0 – 7FFFFFFF h
Default Value	256

Index	6066h
Name	Following_error_time_out
Object Code	VAR
Data Type	UINT16
Access	RW
PDO Mapping	YES
Units	ms
Value Range	0 – 65535
Default Value	0

Index	60FAh
Name	Control effort
Object Code	VAR
Data Type	INT32
Access	RO
PDO Mapping	YES
Units	Speed units
Value Range	----
Default Value	----

Index	6067h
Name	Position window
Object Code	VAR
Data Type	UINT32
Access	RW
PDO Mapping	YES
Units	Position units
Value Range	----
Default Value	400

Index	6068h
Name	Position_time
Object Code	VAR
Data Type	UINT16
Access	RW
PDO Mapping	YES
Units	ms
Value Range	0 – 65535
Default Value	0

9. Periodic synchronization position mode

The periodic synchronous position mode is used in Ether CAT communication, that is, the working mode is set to 9, but in some cases, for the convenience of users, the response mechanism of the profile position mode is omitted. Therefore, this mode is also supported on the can bus. .

Using the periodic synchronous position mode, the upper control will execute as many positions as the given position, and there is no need for the response mechanism like the PP mode for position control.

During operation, this mode can also adjust the maximum speed of the running motor as needed (6080_00, data type UINT32)..

10. Application examples

10.1 Control speed loop

1) Write 3 to Operation Mode (index 0x6060), indicating that the speed loop is selected, and the current working mode can be observed through Display Mode (index 0x6061);

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4
601	2F	60	60	00	03

2). If the lower 4 bits of the current control word Control Word are 0, write 6, 7, and 15 to the lower 4 bits of the control word in turn, indicating that it is enabled.

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5
601	2B	40	60	00	06	00

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5
601	2B	40	60	00	07	00

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5
601	2B	40	60	00	0F	00

(3) Write the speed to be controlled to Target Velocity (index 0x60FF), unit: default RPM. The current motor speed feedback value can be observed through VelocityActualValue (index 0x6069), and the given speed can be observed through VelocityDemandValue (index 0x606B).

The host computer device sends, writes to the drive at a speed of 200RPM

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	23	FF	60	00	C8	00	00	00

Speed mode

1.Operating mode 16#6060

Value Description

- -128,…,-2 Reserved
- -1 No mode
- 0 Reserved
- 1 Profile position mode
- 2 Velocity (not supported)
- **3 Profiled velocity mode**
- 4 Torque profiled mode
- 5 Reserved
- 6 Homing mode
- 7 Interpolated position mode
- 8…127 Reserved

2.Status word 16#6041

3.Control word 16#6040

Process control

16#0006

↓

16#0007

↓

16#000F

↓

Enable

↓

Given by speed 16#60FF

10.2 Control position loop

10.2.1 Absolute or relative position control mode

(1) Write 1 to the operating mode Operation Mode (index 0x6060), indicating that the profile position loop is selected, and the current operating mode can be observed through Display Mode (index 0x6061);

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4
601	2F	60	60	00	01

(2) This parameter Profile Velocity (index 0x6081) controls the speed of the position loop, which represents the final speed after the position is started and the acceleration is completed;

The upper computer device sends and writes to the drive at a speed of 800RPM

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	23	81	60	00	20	03	00	00

(3) If the lower 4 bits of the current control word Control Word are 0, write 6, 7, 15 to the lower 4 bits of the control word in turn, indicating

Enable.

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5
601	2B	40	60	00	06	00

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5
601	2B	40	60	00	07	00

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5
601	2B	40	60	00	0F	00

(4) Write the target position to be turned to Target Position (index 0x607A), unit: default pulse number, you can observe the current motor position feedback value through Positioned (index 0x6064),

The upper computer equipment sends, assuming that the absolute position is given 10000 pulses

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	23	7A	60	00	10	27	00	00

(5) Setting the control word ControlWordbit4 to 1 means setting a new position. The host computer reads the status word bit12 as 1 which means that after the response is successful, the control word bit4 is cleared to 0, and then the drive starts to execute planning.

The upper computer equipment sends, assuming that the absolute position control is used, and it is executed immediately,

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5
601	2B	40	60	00	1F	00

10.2.2 Function description

There are two ways to give a target location:

Single step setting:

After the motor reaches the target position, the driver informs the host of "target position reached", and then acquires a new target position and starts to move. Before acquiring a new target position, the motor speed is usually zero.

Continuous setting:

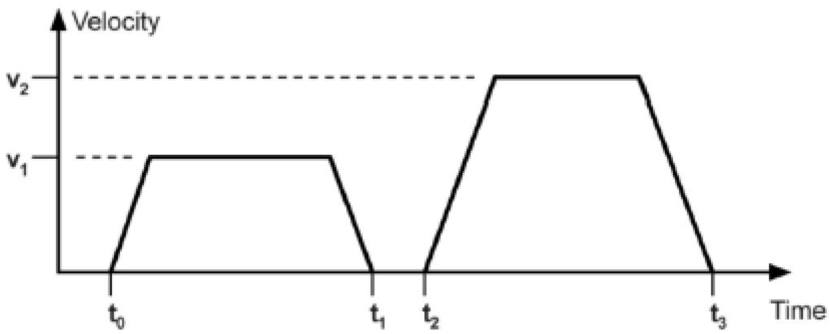
After the motor reaches the target position, it immediately continues to move towards the next preset target position. In this way, the effect of continuous motion without pause can be achieved. Between the two target positions, the motor does not need to decelerate to zero

The above two methods can be changed in real time by bit4 and bit5 of the control word and bit12 (set_point_acknowledge) of the status word.

Through the handshake mechanism, the position control that is being executed can be interrupted, and the target position can be reset and the execution can be started by using these few words.

Steps of the single-step setting method:

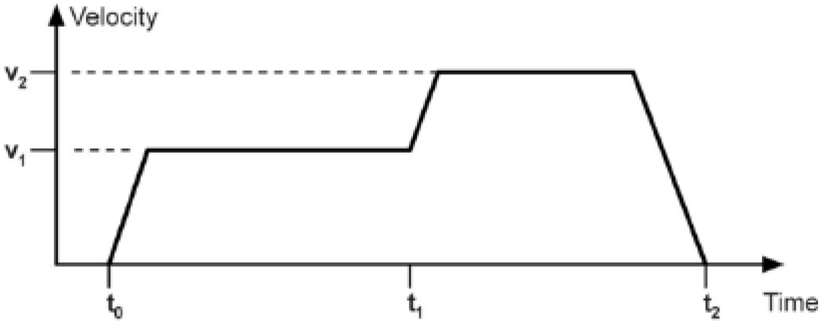
First set the network management (NMT) status to "Operational", and set the control mode parameter (6060 h) to 1.
1. Set the given target position (target_ position: 607A h) and other parameters according to actual needs;
2. Set bit4 (new_set_point) of the control word control word to "1"; bit5
(Change_set_immediately) set to "0"; bit6
(Absolute/Relative) is determined by the target position type (absolute or relative);
3. Set the drive response in bit12 (set_point_acknowledge) of status word, then
Then start to perform position control;
4. After reaching the target position, the drive responds via bit10 (target reached) of the status word.
Then continue to move according to the procedure or accept a new target position.



Steps of continuous setting method:

First set the network management (NMT) status to "Operational", and set the control mode parameter (6060 h) to 1.
1. Set the first given target position (target_ position: 607A h), target speed, acceleration/deceleration and other related parameters according to actual needs;
2. Set bit4 (new_set_point) of the control word control word to "1"; bit5
(Change_set_immediately) is set to "0"; bit6 (absolute/relative) is determined by the target position type (absolute or relative);
3. Set the driver response in bit12 (set_point_acknowledge) of status word, and then start to execute position control;
4. Set the second given target position (target_ position: 607A h), target speed, acceleration/deceleration and other

related parameters;
5. Set bit4 (new_set_point) of the control word control word to "1"; bit5
(Change_set_immediately) is set to "0"; bit6 (absolute/relative) is determined by the target position type (absolute or relative);
6. After reaching the first target position, the driver will continue to control the second target position without stopping; when the second target position is reached, the driver will respond via bit10 (target reached) of the status word status word. Then continue to move according to the procedure or accept a new target position.



Notice:
1. The drive is powered on by default acceleration and other parameters. If the user does not set the acceleration, it will be controlled by the default parameters. If the acceleration needs to be set,
It must be set after the current position is completed, and the setting is invalid after running. If you go to continuous positions and only set one speed, then the following positions will be subject to the first speed.

Location mode

1. Operating mode16#6060

Value Description

-128…-2 Reserved

-1 No mode

0 Reserved

1 Profile position mode

2 Velocity (not supported)

3 Profiled velocity mode

4 Torque profiled mode

5 Reserved

6 Homing mode

7 Interpolated position mode

8…127 Reserved

2. Status word 16#6041

3. Control word 16#6040

Process control

16＃0006

↓

16＃0007

↓

16＃000F

↓

Enable

↓

Write destination 16＃607A (如 100000)

↓

16＃001F

10.3 Control current loop

(1) Write 4 to Operation Mode (index 0x6060), indicating that the current loop is selected, and the current working mode can be observed through Display Mode (index 0x6061);

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4
601	2F	60	60	00	04

(2). If the lower 4 bits of the current control word are 0, write 6, 7, 15 to the lower 4 bits of the control word in turn, indicating that the enable is enabled.

Host computer equipment sending

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5
601	2B	40	60	00	06	00

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5
601	2B	40	60	00	07	00

Host computer equipment sending

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5
601	2B	40	60	00	0F	00

(3) Write the torque value to be controlled to Target Torque (index 0x6071), unit: 0.1 ampere by default. You can use Torque Actual Value (index 0x6077) to observe the current torque feedback value of the motor, and you can use CurrentFdb

(Index 0x6078) Observe the current feedback value of the motor.

Note: The current loop mode runs in full speed operation mode.

The upper computer equipment sends, given a current of 2 amperes

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5
601	2B	71	60	00	14	00

1. Operating mode 16#6060

Value Description

-128…-2 Reserved

-1 No mode

0 Reserved

1 Profile position mode

2 Velocity (not supported)

3 Profiled velocity mode

4 Torque profiled mode

5 Reserved

6 Homing mode

7 Interpolated position mode

8…127 Reserved

2. Status word 16#6041

3. Control word 16#6040

Process control

16#0006

↓

16#0007

↓

Run
16#000F

↓

Enable

↓

Write target torque 16#6071 For example, 50 is 5%

↓

10.4 Parameter save

After modifying the parameters, if the user wants the parameters to be valid after power-on again, they can write 0x65766173 to Save Para (index 0x1010, sub-index 01). After the driver receives the command, it will save the driver parameters to the external memory. In, the storage time takes about 2-3 seconds.

Name	Index (H)	Subindex	Attribute	Data type
Save parameters	1010	01	RW	RW

For example, send via SDO, assuming the node id is 0x01

COBID	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
601	22	10	10	01	73	61	76	65

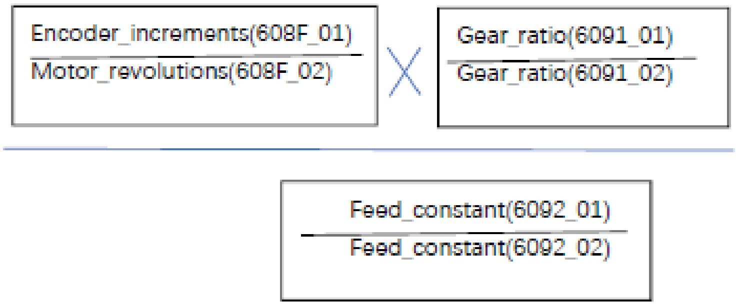
10.5. Set electronic gear ratio

The default electronic gear ratio is 1. If you need to change it, you can configure it through can or through the driver's own debugging software. If the current electronic gear ratio is 2, the position setting (607A_00) is given 1000 pulses. After running, the motor It will actually go to 2000 pulse positions.

The configuration of electronic gear ratio will affect key control parameters such as position setting, position feedback, speed setting, speed feedback, etc.

The drive calculates the gear ratio by indexing these objects: 608F,, 6091,, 6092

Electronic gear ratio=



Parameters affected by electronic gear ratio:

Position actual value (6064_00),

Target Position(607A_00),

Velocity Demand Value(606B_00),

Velocity Actual Value(606C_00),

Profile Velocity (6081_00),

Target Velocity(60FF_00) and other parameters.

10.6 Position clear function

There are many ways to eliminate position feedback errors, here only the commonly used ones are introduced:

(1) If it is detected that bit13 of the control word (0x6040_00) is set to 1, the position feedback will continue to be cleared to 0, and at the same time, bit14 of the status word (0x6041_00) is also set to 1 as a response.

(2). Clear the position feedback through the shortcut command 1

Index	Subindex	meaning	Annotation
0x2004	0x20	Shortcut instructions	Perform different operations based on different values Command, reset to 0 after each write 0: invalid command 1: Fault reset 2: Elimination of position error (assign position feedback value to position reference) 3: Elimination of position error (reset position feedback value to 0) 4: Elimination of position error (reset both position feedback value and position reference value to 0)

(3) Use the zero return mode 35 to set the current point as the origin to realize the position feedback reset.

Appendix

Object Dictionary--- CiA301 device profile

Index/Sub Index	Name	Data Type	Attribute	Mappable	Comment
1000/00	Device Type	UINT32	RO	NO	
1001/00	Error Register	UINT8	RO	NO	
1008/00	Device Name	UINT32	RO	NO	
1009/00	Hardware Version	UINT32	RO	NO	
100A/00	Software Version	UINT32	RO	NO	
1010/01	Save Para	UINT32	Sub0:RO Sub1:RW	NO	
1015/00	Emcy Inhibit Time	UINT16	RW	NO	
1016/01	Consumer Heart Time	UINT32	Sub0:RO Sub1:RW	NO	
1017/00	Product Heart Time	UINT16	RW	NO	
1018/04	Identity Object	UINT32	RW	NO	
1400/01-1403/01	RPDO_COB_ID	UINT32	RW	NO	
1400/02-1403/02	RPDO_Trans Type	UINT8	RW	NO	
1600 1601 1602 1603	RPDO mapping parameters	UINT32	RW	NO	PDO mapping parameters Sub1-Sub4
1800/01-1803/01	TPDO_COB_ID	UINT32	RW	NO	
1A00 1A01 1A02 1A03	TPDO mapping parameters	UINT32	RW	NO	PDO mapping parameters Sub1-Sub4
200B/01	Digital IO input status	UINT16	RO	TxMap	
200B/02	Digital IO output status	UINT16	RO	TxMap	
200B/03	Real-time failure 1	UINT16	RO	TxMap	
200B/04	Real-time status 1	UINT16	RO	TxMap	
200B/05	Current feedback	INT16	RO	TxMap	0.1 amp
200B/06	Internal speed feedback	INT32	RO	TxMap	Default RPM
200B/07	Internal position feedback	INT32	RO	TxMap	

200B/0C	Drive heat sink temperature	INT16	RO	TxMap	Celsius
200B/0D	Motor temperature	INT16	RO	TxMap	Celsius
200B/0E	Drive CPU temperature	INT16	RO	TxMap	Celsius
200B/10	bus voltage	UINT16	RO	TxMap	0.1 volt
200B/11	Analog input 0	INT16	RO	TxMap	Millivolt (mV)
200B/11	Analog input 1	INT16	RO	TxMap	Millivolt (mV)
200B/1C	Real-time fault 2	UINT16	RO	TxMap	
200B/1D	Real-time status 2	UINT16	RO	TxMap	
200B/26	Current enable duration	UINT32	RO	TxMap	0.1 second
200B/27	Last enable duration	UINT32	RO	TxMap	0.1 second

Object dictionary --- DS402 device profile

Index/Sub Index	Name	Data Type	Attribute	Mappable?	Comment
603F/00	Error Code	UINT8	RO	TxMap	
6040/00	Control Word	UINT16	RW	RxMap	
6041/00	Status Word	UINT16	RO	TxMap	
6502/00	supported_drive_mode	UINT32	RO	NO	
605A/00	Quick_stop_option_code	INT16	RW	NO	
605B/00	Shutdown_option_code	INT16	RW	NO	
605C/00	Disable_operation_option_code	INT16	RW	NO	
605D/00	Halt_option_code	INT16	RW	NO	
605E/00	Fault_reaction_option_code	INT16	RW	NO	
6060/00	Operation Mode	INT8	RW	RxMap	
6061/00	Display Mode	INT8	RO	TxMap	
6062/00	Position demand value	INT32	RO	TxMap	
6063/00	position_actual_internal_value	INT32	RO	TxMap	
6064/00	Position actual value	INT32	RO	TxMap	
6067/00	Reach Position	UINT32	RW	NO	
6069/00	Velocity Sensor Actual Value	INT32	RO	TxMap	
606A/00	Sensor Select	INT16	RO	NO	
606C/00	Velocity Actual Value	INT32	RO	TxMap	
606D/00	Velocity window	UINT16	RW	NO	

606E/00	Velocity window time	UINT16	RW	NO	
606F/00	Velocity threshold	UINT16	RW	NO	
6070/00	Velocity threshold time	UINT16	RW	NO	
6071/00	Target Torque	INT16	RW	RxMap	
6072/00	Max_torque	INT16	RW	NO	
6073/00	Max Currentt	INT16	RW	NO	
6074/00	torque_demand	INT16	RO	TxMap	
6075/00	Motor_rated_current	UINT32	RW	NO	
6076/00	Motor_rated_torque	UINT32	RW	NO	
6077/00	TorqueActualValue	INT16	RO	TxMap	
6078/00	Currentt ActualValue	INT16	RO	TxMap	
607A/00	Position Target value	INT32	RW	RxMap	
607B/00	position_range_limit	INT32	RW	RxMap	ARRAY
607C/00	Home Offset	INT32	RW	NO	
607D/00	Software_position_limit	INT32	RW	RxMap	ARRAY
607E/00	Polarity	UINT8	RW	FALSE	
607F/00	max_profile_velocity	UINT32	RW	NO	
6080/00	max_motor_speed	UINT32	RW	NO	
6081/00	Profile Velocity	UINT32	RW	RxMap	
6082/00	End Velocity	UINT32	RW	RxMap	
6083/00	Profile acceleration	UINT32	RW	RxMap	
6084/00	profile_deceleration	UINT32	RW	RxMap	
6085/00	quick_stop_deceleration	UINT32	RW	RxMap	
6086/00	motion_profile_type	IINT16	RW	NO	
6087/00	torque_slope	UINT32	RW	NO	
608F/02	Position encoder resolution	UINT32	Sub 0:RO Sub 1,2:RW	NO	
6093/02	Position factor	UINT32	Sub 0:RO Sub 1,2:RW	NO	
6094/02	Velocity encoder factor	UINT32	Sub 0:RO Sub 1,2:RW	NO	
6095/02	Velocity factor 1	UINT32	Sub 0:RO	NO	

			Sub 1,2:RW		
6097/02	Acceleration factor	UINT32	Sub 0:RO Sub 1,2:RW	NO	
6098/00	Homing Method	INT8	RW	NO	
6099/01	High Homing Speed	INT32	RW	NO	
6099/02	Low Homing Speed	INT32	RW	NO	
60B1/00	velocity_offset	INT32	RW	NO	
60B2/00	torque_offset	INT16	RW	NO	
60F2/00	Positioning_option_code	UINT16	RW	NO	
60FD/00	Digital_Inputs	UINT32	RO	NO	
60FE/00	Digital_Outputs	UINT32	RO	NO	
60FF/00	Target Velocity	INT32	RW	RxMap	

Object Dictionary---Introduction to CiA301 Communication

Objects 0x1400 - 0x1403: *Receive PDO communication parameter*

- Object description:

Index	1400h - 1403h
Name	Receive PDO Parameter
Object code	RECORD
Data type	PDO CommPar (object 0x20)
Category	Conditional: mandatory for each supported PDO

- Entry description:

Sub-index	0
Description	Number of entries
Entry category	Optional
Access	Read only
PDO mapping	No
Value range	UNSIGNED8
Default value	2

Sub-index	1
Description	COB-ID used by PDO
Entry category	Optional
Access	Read only
PDO mapping	No
Value range	UNSIGNED32
Default value	Index 1400h: 0x27F Index 1401h: 0x37F Index 1402h: 0x47F Index 1403h: 0x57F

Sub-index	2
Description	Transmission type
Entry category	Optional
Access	Read only
PDO mapping	No
Value range	UNSIGNED8
Default value	255

Objects 0x1800 - 0x1803: *Transmit PDO communication parameter*

- Object description:

Index	1800h - 1803h
Name	Transmit PDO parameter
Object code	RECORD
Data type	PDO CommPar (object 0x20)
Category	Conditional: mandatory for each supported PDO

- Entry description:

Sub-index	**0**
Description	Number of entries
Entry category	Optional
Access	Read only
PDO mapping	No
Value range	UNSIGNED8
Default value	No

Sub-index	**1**
Description	COB-ID used by PDO
Entry category	Optional
Access	Object 1800h: Read/Write Object 1801h: Read/Write Object 1802h: Read/Write Object 1803h: Read/Write
PDO mapping	No
Value range	UNSIGNED32
Default value	Index 1800h: 0x400001FF Index 1801h: 0x400002FF Index 1802h: 0x400003FF Index 1803h: 0x400004FF

Sub-index	**2**
Description	Transmission type
Entry category	Optional
Access	Object 1800h: Read/Write Object 1801h: Read/Write Object 1802h: Read/Write Object 1803h: Read/Write
PDO mapping	No
Value range	0…240
Default value	0

Sub-index	**3**
Description	Inhibit time
Entry category	Optional
Access	Read/Write
PDO mapping	No
Value range	UNSIGNED16
Default value	0 (no inhibit time between messages)

Sub-index	**4**
Description	Reserved
Entry category	
Access	
PDO mapping	
Value range	
Default value	

Sub-index	**5**
Description	Event timer
Entry category	Optional
Access	Read/Write
PDO mapping	No
Value range	0 - not used UNSIGNED16
Default value	No